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MILITARY SPECIFICATION

SUPPRESSION, RADIO INTERFERENCE GENERAL REQUIREMENTS FOR VEHICLES (AND VEHICULAR SUB-ASSEMBLIES)

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 This specification covers the requirements for radio interference suppression, within the frequency range of 0.15 to 1000 megacycles, for all internal combustion engine-driven vehicles, tactical and nontactical procured by the Ordnance Corps for all the services, to permit satisfactory reception of intelligence with high sensitivity electronic communication equipment installed as follows:

In the vehicle and connected to the vehicular battery or auxiliary power supply installed in the vehicle as an integral part thereof.
In the vehicle and connected to a separate power source.
Adjacent to the vehicle and connected to a separate power source.

2. APPLICABLE SPECIFICATIONS AND DRAWINGS

2.1 Specifications.—The following specifications, of the issue in effect on date of invitation for bids, form a part of this specification:

MILITARY SPECIFICATIONS

JAN-C-5 —Capacitors, mica-dielectric, fixed.
JAN-C-25 —Capacitors, Direct-Current, Paper-Dielectric, Fixed (Hermetically Sealed in Metallic Cases).
JAN-C-91 —Capacitors, Paper Dielectric, Fixed (Non-Metallic Cases).

U. S. ARMY SPECIFICATIONS

71-1585 —Resistors - Suppressors, Suppression (Radio-Frequency Interference).
71-1667 —Capacitors, By-Pass, Suppression (Radio Frequency Interference).
71-1668 —Filters, Suppression (Radio Frequency Interference).

(*Army.*—Copies of specifications should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(*Navy.*—Copies of Military specifications may be obtained upon application to the Bureau of Supplies and Accounts, Navy Department, Washington 25, D. C., except that activities of the Armed Forces should make application to the Commanding Officer, Naval Supply Depot, Scotia 2, N. Y. Both the title and identifying number or symbol should be stipulated when requesting copies.)

(*Air Force.*—Copies of Military, and U. S. Air Force specifications may be obtained upon application to the Commanding General, Air Development Force, Wright-Patterson Air Force Base, Dayton, Ohio. Both the title and identifying number or symbol should be stipulated when requesting copies.)

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2.2 **Drawings.**—The following drawings, of the issue in effect on the date of invitation for bids, form a part of this specification:

SC–D–28119—Typical Non-Tactical Vehicle Radio Interference Suppression System (Battery Ignition).

SC–D–28120—Typical Tactical Vehicle Suppression System.

(Copies of Signal Corps Drawings may be obtained from the Signal Corps Procurement Agency, 225 South 18th Street, Philadelphia 3, Pa.)

3. REQUIREMENTS

3.1 Definitions.

3.1.1 *Radio interference.* — "Radio interference," or simply "interference," is defined as any electrical disturbance which causes undesirable response or malfunctioning of communication equipment. Interference emanating from a source may reach the affected equipment by means of conduction, induction, radiation, or any combination thereof.

3.1.2 *Units for measurement of intensity of interference.*—Intensity of interference is measured, for the purposes of this specification, in terms of microvolt per kilocycle (kc.) or per megacycle (mc.) of bandwidth (uv/kc or uv/mc). Interference intensity in microvolts per kc. or mc. is equal to the number of r.m.s. sinewave microvolts (unmodulated), applied to the input of the measuring circuit at its center frequency, which will result in peak response in the circuit equal to that resulting from the interference pulse being measured, divided by the effective bandwidth of the circuit in kilocycles or megacycles. The effective bandwidth is the area divided by the height, of the voltage-response-versus-radio-frequency, selectivity curve, from antenna to peak detector.

3.1.3 *Suppression.*—"Suppression" of radio interference is reduction of the intensity of the emanated interference by means applied to or at the source.

3.1.4 *Suppression system.* — All components, materials and their application which are used for suppression of the interference emanated by a vehicle will be referred to as the "suppression system."

3.1.5 *Electrical subassemblies.* — An electrical subassembly is defined for purposes of this specification as any vehicular electrical subassembly such as spark plugs, generator, regulator, etc., associated with the basic functioning of the vehicle or its electrical system, which is supplied, stored, and issued as a complete operating assembly.

3.1.6 *Electrical accessory units.*—An electrical accessory unit is defined for the purposes of this specification as any electric motor or electro-mechanical device, such as gun traverse motors, cant correctors, synchro-motors, relay mechanisms, etc., capable of operation independent of the vehicle or its electrical subassemblies except for the power source.

3.1.7 *Type approval.* — An approved-type suppression component is one for which samples of the same make and type have previously been submitted, and tested and approved as being suitable for suppression purposes and in accordance with the applicable subsidiary specification.

3.1.8 *Integral suppression.*—An integrally suppressed electrical subassembly or accessory unit is one in which the suppression components are procured, stored, and issued as integral parts thereof.

3.2 Application of specification.

3.2.1 *Tactical vehicles.* — The following types of vehicles shall conform to this specification for tactical vehicles; unless otherwise specified, all types of self-propelled vehicles which receive installations of communication equipment, all types of tanks, armoured cars ,self-propelled cannon, self-propelled cargo carriers, half-track personnel carriers, half-track cargo vehicles, motor transport vehicles, high-speed tractors, and motorcycles.

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3.2.2 *Nontactical vehicles.* — The following types of vehicles shall conform to this specification for nontactical vehicles, unless otherwise stated in the bid request and contract, in which case they shall meet the requirements for tactical vehicles: passenger-type vehicles and buses.

3.2.3 *Self-propelled machinery.*—Self-propelled machinery such as excavators, road graders, fork-lift trucks, slow-speed tractors, etc, shall not be considered as vehicles within the scope of this specification unless otherwise stated in the primary equipment specification. In this case, the latter specification shall state whether the requirements for tactical or nontactical vehicles shall apply.

3.2.4 *Railroad equipment.*—This specification shall apply to all railroad equipment as specified in the primary equipment specification or in the bid request and contract. (See 6.11.)

3.3 Performance (tactical vehicles).—The suppression system shall effectively reduce all radio interference emanating from the vehicle, to a level which would not prevent satisfactory reception of a signal which is just strong enough to be easily intelligible, in the absence of interference, on high-sensitivity radio receiving equipment in the required frequency range. This requirement applied to interference generated by the electrical system of the vehicle, auxiliary equipment, accessories, intermittent contact between parts, electrostatic discharge, or any other source which may appear.

3.3.1 *Limits of interference for tactical vehicles.*—Tactical vehicles not emanating interference in excess of the limits stated in 3.3.1.1, 3.3.1.2, and 3.3.1.3 when tested according to this specification, will be considered satisfactory in regard to interference emanation.

3.3.1.1 *Outside radiated interference limits.*—The interference limits stated in table I apply to tactical vehicles and accessories. (See 4.3 and 4.5.)

3.3.1.2 *Inside radiated interference limits.*—The interference limits stated in table II apply to the level of radiated interference permitted inside turreted vehicles being tested under 4.3, 4.3.2 to 4.3.2.2, and 4.5.

TABLE I.—*Outside radiated interference limits.*

Test equipment	Frequency band or tuning unit	Frequency range mc.	Limit	
			uv/unit bandwidth	db above 1 uv/mc.
Test Set AN/URM-3 R.F. Interference Test Set AN/URM-29		0.15 to 40.0	.75 uv/kc.	58
	TN-16/APR-4	40 to 95	100 uv/mc.	40
	TN-17/APR-4	96 to 305	200 uv/mc.	46
	TN-187/URM-29	306 to 1000	200 uv/mc.	46

TABLE II.—*Inside radiated interference limits.*

Test equipment	Frequency band or tuning unit	Frequency range mc.	Limit	
			uv/unit bandwidth	db above 1 uv/mc.
Test Set AN/URM-3 R.F. Interference Test Set AN/URM-29		1.5 to 40	.25 uv/kc.	48
	TN-16/APR-4	40 to 95	100 uv/mc.	40
	TN-17/APR-4	96 to 305	200 uv/mc.	46
	TN-187/URM-29	306 to 1000	200 uv/mc.	46

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3.3.1.3 *Conducted interference limits at terminal boxes.* — The interference limits stated in table III apply to the level of conducted interference between any pair of terminals intended to supply power to communications equipment at any electrical terminal box or contact block in tactical vehicles being tested under 4.3, 4.3.3 to 4.3.3.1, 4.4 and 4.5, with the engine and all electrical equipment in operation.

3.4 Requirements (nontactical vehicles).— All nontactical vehicles shall incorporate, as a minimum requirement, the suppression system detailed in Drawing SC–D–28119.

3.4.1 Nontactical vehicles which incorporate any interference-producing device not covered by the minimum suppression system shown on Drawing SC–D–28119 shall be suppressed, with all interference sources operating, to the level of interference which would exist if the vehicle did not incorporate such device. The contracting officer may run tests to determine conformance to this requirement.

3.5 Approval of suppression components (all vehicles).—All components of the suppression system for both tactical and nontactical vehicles shall be subject to approval as hereinafter provided.

3.5.1 *Integral suppression components.*— Integral suppression components (see 3.1.8), shall be subject to approval of the particular application in the sub-assembly or accessory for soundness of electrical, mechanical and suppression features. Whenever practicable, such components shall be in accordance with any of the applicable Military or U. S. Army specifications listed in 2.1 of this specification, and shall have type approval as provided in 3.5.2. Whenever components complying with these specifications cannot be used, the specific component used shall be subject to approval for the specific application.

3.5.2 *Nonintegral suppression components.* —Nonintegral suppression components shall be in accordance with Specifications 71–1585, 71–1667 or 71–1668, and shall have type approval. (See 3.1.7) If the bidder or contractor proposes to use any component not previously approved, samples of such components shall be submitted for approval, as described in 6.2. Capacitors conforming to Specifications Jan–C–5, JAN–C–25 or JAN–C–91 are not necessarily suitable for suppression purposes, but specific applications may be approved.

3.5.3 *Use of suppression components beyond specification ratings.*—Capacitors, filters or resistor-suppressors shall not be used under conditions more severe than the conditions specified in the applicable subsidiary specifications, unless specifically authorized in writing by the contracting officer. Such authorization may be obtained only by submission of samples, as described in 3.5.2, accompanied by a complete statement of the service conditions under which the components will be used, and test data showing that the component will withstand such conditions.

TABLE III.—*Conducted interference limits at terminal boxes.*

Test equipment	Coupler unit for connection to terminal box	Frequency range mc.	Limit	
			uv/unit bandwidth	db above 1 uv/mc.
Test Set AN/URM–3	CU–152/URM	1.5 to 10 10 to 40	10/uv/kc. 5/uv/kc.	80 74

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3.5.4 *Suppression components not covered by subsidiary specifications.*

3.5.4.1 Approval of the use of any capacitor, filter, filter assembly, or resistor-suppressor or suppression, other than those covered by 3.5 to 3.5.3, inclusive, will be granted by the contracting officer only if the contractor show that suppression or electrical requirements, operating conditions, or limitations of space prevent the use of types in accordance with the applicable subsidiary specifications. In such cases, samples shall be submitted to the contracting officer in accordance with 3.5.4.3.

3.5.4.2 At the discretion of the contracting officer, suppression components other than those covered by 3.5 to 3.5.3, inclusive such as shields and shielding, lockwashers and bonds, shall be subject to inspection for mechanical, electrical and suppression effectiveness before approval for use. Samples shall be submitted for the purpose in accordance with 3.5.4.3.

3.5.4.3 *Submission of samples for inspection, test and approval.*—A maximum of 12 samples shall be submitted of any component covered by 3.5.4.1 and 3.5.4.2. These samples shall be constructed using the materials, parts, tools, and methods which will be used in quantity production. When submitted, they shall be acompanied by five copies of detail drawings showing dimensions, materials, and construction, a complete statement of the service conditions under which the component will be used, and test data showing that the component will withstand such conditions.

3.6 Integrally suppressed electrical subassemblies and accessory units.—Integral suppression of electrical subassemblies and accessories shall be used whenever practical, in order to improve suppression, simplify the maintenance of the suppression system, and reduce the chances that it may unintentionally be rendered ineffective during the course of vehicle maintenance.

3.7 Pilot model (tactical vehicles).

3.7.1 *Submission of pilot model.*—The contractor shall submit an experimental pilot model of each vehicle to the bureau or agency designated in the bid request and contract, for inspection and test of the suppression system. The experimental pilot model shall be a preproduction vehicle, and may be hand-built. The contractor shall equip this model with suppression components in the same manner that he intends to equip production vehicles. Before submission for approval, he shall test it for conformance with this specification. (See 6.8.) The first experimental pilot model shall be accompanied by two copies of each of the following:

Detailed list and description of the suppression components, integral and nonintegral, indicating for each suppression component the manufacturer's name and model number, and location on the vehicle or in the subassembly or accessory.

List of all electrical equipment, except lights and horn, used on vehicle. Each such item of electrical equipment shall be identified on the list by manufacturer's name and model number.

Print of the complete electrical system of the suppressed vehicle, showing electrical location and connections of all suppression components, whether integral or nonintegral.

Each of the above items shall be marked with the date of submission of vehicle, contract number, name and address of the contractor, the nomenclature of the vehicle, and its serial and U. S. registration numbers.

3.7.2 *Approval of pilot model.*—The pilot model will be inspected and tested by the designated Government agency for compliance with this specification (see 6.4), and will then be returned to the contractor. If the pilot model is disapproved, the contractor shall correct the deficiencies and submit it, unless otherwise allowed, until approval is obtained. Each corrected model shall be ac-

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companied by a complete description of the changes made to correct the faults of the preceding rejected model. Approval of the pilot model shall not constitute a waiver of any specified requirement.

3.7.3 *Report on pilot model.*—The Government testing agency will furnish the contracting officer a report on the approved pilot model, setting forth the detailed suppression requirements to be followed on production vehicles and accessories; however, if subsequent tests on production vehicles indicate that these requirements must be modified to provide a satisfactory suppression system, the modified requirements shall be followed thereafter. The contracting officer will furnish the contractor with instructions, based on the report of the approved pilot model, detailing the suppression system to be supplied on production vehicles. Authority in writing shall be obtained from the contracting officer for any exceptions to, or deviations from, the suppression system approved for production.

3.7.4 *Approval of electrical accessory units and subassemblies.*

3.7.4.1 *Accessories.*—A model of each noncommercial type electrical accessory as defined in 3.1.6 (or redesigned commercial type) designed by the contractor or his sub-contractor for the particular vehicle on the contract, and not approved since the date of this specification for use on Army vehicles, shall be submitted, for inspection and test of its suppression system (including all pertinent nonintegral suppression components), prior to submission of the experimental pilot model of the vehicle, to the agency designated in the bid request and contract. The model of the accessory shall be accompanied by two copies of a detailed list and description of the suppression components, both integral and nonintegral, indicating for each suppression component the manufacturer's name and model number, and location in or relative to the accessory.

3.7.4.2 *Subassemblies.*—No pilot testing of vehicular electrical subassemblies as defined in 3.1.5 shall be required under this specification, as suppression tests will be performed on the pilot and production models of the vehicles incorporating the subassemblies.

3.7.5 *Changes after approval of pilot model.*—After approval of the pilot model all production vehicles, accessories, and subassemblies shall unless otherwise allowed or required, be identical in all respects to the approved pilot model and the report on that model (3.7.3) except as described below:

3.7.5.1 *Substitution of suppression components.*—

5.7.5.1.1 The contractor may, at his discretion, substitute a suppression component with identical electrical characteristics and mechanically interchangeable construction, but of different manufacture, for a corresponding component, provided that the substitute component has received type approval as described in 3.5 to 3.5.3, inclusive.

3.7.5.1.2 Any other substitution or change of a suppression component shall be made only if approved in writing by the contracting officer. Request for permission to make such a substitution or change shall be accompanied by a detailed description of the proposed substitute suppression component and its method of application, and shall include data showing that the proposed substitute is at least the equal of the item it will replace. (If the proposed substitute is an item requiring type approval under 3.5 and has not yet received such approval, the contractor shall submit samples in accordance with that paragraph.) At the discretion of the contracting officer, the contractor shall submit another pilot model, identical with the previously approved pilot model except for the changes being submitted for approval.

3.7.5.1.3 *Changes in integral components.*—In the case of substitutions or changes of integral suppression components incorpora-

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ted in electrical subassemblies or accessory units, the contracting officer may require submittal of a model of the original and of the revised subassembly or accessory unit, and if the revised version is found inferior as to suppression characteristics, the contracting officer may then require test of an additional pilot model of the vehicle incorporating the revised subassembly or accessory unit.

3.7.5.2 *Changes in the vehicle (tactical only).*—The contracting officer shall be notified as early as possible in writing by the contractor of any changes made or planned in the electrical components, subassemblies or accessories of the vehicle, their location or wiring. He shall likewise be notified of any changes in the structure or electrical bonding (including use of tooth-type lockwashers) of the vehicle which might affect generation or emanation of interference. If the contractor does not submit data satisfactory to the contracting officer showing that the changes will not result in failure to conform to this specification, the contracting officer may, at his discretion, require the contractor to submit another pilot model, identical with the previously approved pilot model except for the changes being submitted for approval. This pilot model shall meet the same requirements as the original pilot model.

3.8 Spare parts.

3.8.1 *Suppression components.* — Spare suppression components for all vehicles shall conform to 3.10 for interchangeability with corresponding parts used on the vehicle, and shall also conform to 3.5.

3.8.2 *Other components.* — Where components of the vehicle, such as fenders and radiators, which are not part of the suppression system, have been modified for suppression purposes by preparing surfaces for bonding or by any other means, the corresponding spare components shall be similarly modified to provide an equal degree of suppression.

3.9 Methods of suppression.—Suppression of ignition, charging circuit, and other electrical interference shall be effected by resistor-suppressors, capacitors and/or filter, shielding, proper bonding and grounding of radiating elements, proper routing of wiring, and proper location of parts.

3.9.1 *Filters and capacitors.*—Filters shall be used only where adequate suppression cannot be effected by reasonable application of capacitors.

3.9.2 *Resistor-suppressors.*—Resistor-suppressors shall be used in the high-tension ignition circuits to minimize the suppression requirements of the shielding and critical mantenance thereof.

3.9.3 *Separation of wiring.*—High-tension ignition wiring, primary ignition wiring, and charging-circuit wiring shall not be routed any closer than necessary to each other or combined with auxiliary wiring. Adequate support shall be given all wiring to maintain separation and position. When interference from electrical fuel pumps and other interference-producing devices cannot be suppressed at the source, wiring for such devices, if combined with other wiring, shall be separately shielded unless it can be demonstrated that the interference caused by the auxiliary devices is of such value and frequency that the vehicle meets specification requirements without the shielding. Connections between the several electrical components of the vehicle and the associated bypass capacitors or filters of the suppression system shall be as short as possible, and shall be separated from other wiring wherever practicable.

3.9.4 *Shielding.* — Flexible metal hose, solid-wall conduit and wire braid, utilized as part of the suppression system or as mechanical support of the wiring, shall be terminated and grounded at both ends by means of suitable fittings or clamps. Pigtail connections are not acceptable. Where shield-

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ing terminates at a case or bulkhead, a good electrical contact shall be insured between the metal case or bulkhead and the shielding, and the metal case shall be grounded as specified in 3.9.5. Long lengths of shielding or protecting members shall be clamped at intervals no greater than 2 feet, by clamps which are soldered to, or exert pressure on, the shielding or protecting member, to insure good and permanent radio-frequency electrical grounding and mechanical support. Contacting surfaces of the clamp, shielding, or supporting member, and the part to which the clamp is affixed, shall be treated as specified in 3.11.3. Where continuity of electrical contact along a seam or joint in shielding or fittings is necessary to obtain adequate suppression, the seam or joint shall be corrosion-resistant and mechanically designed to maintain the contact for the required life of the parts, including necessary disassembly and reassembly. Electrically conductive gaskets may be used for this purpose.

3.9.4.1 However, shielding boxes, terminal boxes, junction boxes, and similar parts need not be plated or metal-dipped, provided that continuous-seam contact proves unnecessary for suppression, and provided that each box and its cover, and all conduit terminations, are effectively grounded by the use of tooth-type lockwashers. When threaded conduit fittings are welded to a box, the threaded nipple shall be plated or made of corrosion-resistant metal.

3.9.5 *Grounding.*—All components of the electrical and suppression system and parts of the vehicle which require radio-frequency return paths to ground shall be so grounded that a corrosion-resistant low-impedance electrical connection will be assured throughout the life of the equipment. Metallic guides which support high-tension wiring shall be grounded to meet these requirements. Mechanical parts such as choke, throttle, speedometer and tachometer cables, oil lines, conduit and steering columns, when they project into the field of radio interference, shall be grounded in such a manner as to prevent reradiation that may cause radio interference. Grounding shall be accomplished by bond straps, tooth-type lockwashers, electrically conductive gaskets or other suitable means.

3.9.6 *Bonds.* — Each bonding strap shall have a length and cross-sectional shape and dimensions consistent with best electrical and mechanical practice. All bonding straps shall be installed so that they will not be broken by vibration or stress due to service conditions. Bonding straps shall be mounted, if practicable, in such a manner that they may be readily replaced. Bonds mounted by bolts shall have suitable terminals. Sheet metal screws shall not be used for fastening bonds. At least one end of each bond shall be bolted in order that the vehicle may be readily disassembled without adversely affecting the suppression.

3.9.7 *Machine-finished surfaces.* — Machine-finished surfaces to which suppression components are affixed need not be plated. Such mating surfaces of machine-finished parts shall be free of corrosion, paint, or other foreign material, and shall not be treated in any manner that will impair their electrical conductivity on the mating surfaces.

3.9.8 *Ignition coil.*—In order to minimize suppression treatment and to aid in providing lasting, effective suppression, the ignition coil, unless otherwise allowed by the contracting officer, shall be located close to or combined with the distributor. If separate, it shall be mounted on a bracket, affixed to the engine block, providing good grounding for the case of the coil.

3.10 Interchangeability.—All parts of the suppression system on any vehicle, and all spare parts for the suppression system, shall be electrically and mechanically interchangeable with all corresponding parts on any other vehicles manufactured under the same contract, and wherever possible, with all other vehicles of the same type.

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3.11 Material and workmanship.

3.11.1 *Material.* — The material for each part shall be as specified. The best material commercially available for the purpose, and capable of withstanding the extremes of temperature, humidity and other operating conditions encountered in arctic and tropical service, shall be used when a definite material is not designated.

3.11.1.1 *Insulation material.* — Insulation material which is exposed to the air shall not be subject to rapid combustion or explode from electrical spark or from heating. The use of flammable or explosive varnish is prohibited.

3.11.2 *Soldering.* — Whenever practicable, soldering shall be done with rosin, or rosin and alcohol, flux. If it proves necessary to use any other flux it shall be chemically neutralized and removed after soldering. Soldering shall not be depended upon for mechanical strength of connections. Where flexible shielding, bonds, etc., are soldered, the flexibility of these parts shall be unaffected.

3.11.3 *Corrosion.* — To prevent corrosion, all hardware (bolts, screws, nuts, lockwashers, retaining nuts, elbows, ferrules, etc.) used in the installation of the suppression system shall be plated by an electrolytic or hot-dip process with a metallic material of high electrical conductivity. The finished surfaces shall be capable of withstanding the test specified in 4.6, after which there shall be not more than six corroded areas per square foot, not more than two corroded areas on any part having a total surface of less than ⅓ square foot, nor any corroded area larger than 1/16 inch in diameter. At the discretion of the contracting officer, samples may be required for the corrosion test: (See 6.2).

3.12 Technical manual and instruction book.—Any technical manual or instruction book regarding the vehicle, prepared and furnished as part of the order by which the vehicle is procured, shall contain complete technical information, including photographs and/or drawings, on the functioning and maintenance of the suppression system.

3.13 Workmanship. — All parts shall be manufactured and finished in a thoroughly workmanlike manner, and in accordance with the best commercial practice for arctic and tropical use. All dimensons, except where tolerances are given on drawings, shall be held as close as is consistent with good shop practice. All parts shall fit and operate in a good, workmanlike manner.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Acceptance inspection.—Unless otherwise specified, acceptance inspection shall be conducted by the Government inspector when equipment is submitted for acceptance on contract.

4.1.1 *Scope of acceptance inspection.*—Acceptance inspection shall include the tests of 4.5 and such other inspection and nondestructive testing of material, parts, components and complete suppression systems as are considered necessary by the Government inspector to determine compliance with this specification and its subsidiaries.

4.2 Test equipment for pilot and production tests.

4.2.1 *Measuring equipment.*—Interference measuring equipment as follows, for conducting interference tests on tactical vehicles, will be provided by the contracting officer and used by the Government inspectors:

Test Set AN/URM–3, 0.15-40 megacycles.

R:F. Interference Test Set AN/URM–29, 40-1000 megacycles.

Upon request, the contractor may obtain information which he requires concerning this test equipment and the manner in which it will be used, to enable him to suppress the pilot model to meet the required tests. Limits

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of allowable interference are stated in 3.3.1 through 3.3.1.3, and the units in which the interference shall be measured are defined in 3.1.2.

4.3 Government tests on pilot models.— Pilot models of vehicles and electrical accessories will be tested by the Government at the location specified by the contracting officer. If the vehicle under test is equipped with accessory items such as heaters, fans, windshield wipers, traversing motors and similar equipment, the tests shall also be performed with all of these components operating.

4.3.1 *Radiation Tests (tactical vehicles).* —The radio interference radiated from all tactical vehicles shall be measured on at least three sides, over the frequency range of .15 to 1000 mc. Radiation tests shall be conducted with all electrical equipment in the vehicle in operation.

4.3.1.1 *Test Set AN/URM–3.* — The test limits specified in table I apply. Over the frequency range .15 to 40 mc, using Test Set AN/URM–3, the test equipment shall be located as shown in figures 1, 2, 3 and 4. Whenever Test Set AN/URM–3 is placed on the vehicle, it shall be insulated therefrom by a nonconducting material of adequate size. With Test Set AN URM–3 located on the fender as close as possible to the engine compartment and arranged so that the antenna is situated near the lateral axis of the engine, the 9 foot, rod antenna shall be sloped across the engine compartment so that the tip of the antenna is located in a vertical plane through the edge of the opposite fender or similar outermost extremity of metal, but the mid-point of the antenna shall not be placed closer than 1 foot from the top of the engine compartment. When testing front or rear radiation, depending upon location of engine compartment, Test Set AN/URM–3 shall be located so that the base of the antenna lies in same horizontal plane as the top of the front radiator grill or rear lower edge of top radiator grill. The 9 foot rod antenna shall be sloped across the engine comparment so that the mid-point of the antenna is not more than 24 or less than 12 inches above the engine compartment. If antenna is unobstructed with Test Set AN/URM–3 as close as possible to the vehicle or engine compartment, the antenna shall be sloped so that the mid-point is 12 inches above the engine compartment.

4.3.1.2 *R.f. interference test Set AN/URM–29.*—Over the frequency range of 40 to 1000 mc, the antenna of r.f. interference Test Set AN/URM–29 shall be located as shown in figures 5, 6, 7 and 8. The apex of the cone is located 3 feet from the edge of the engine compartment on the lateral or longitudinal axis of the engine. The antenna configuration used for the various frequency bands and tuning heads is as follows:

(a) 40 to 55 mc. (TN–16/APR–4) Antenna AT–292/URM–29, with 6 Mast Sections AB–21/GR at center of cone

(b) 56 to 95 mc. (TN–16/APR–4) Antenna AT–292/URM–29 with 2 Mast Sections AB–21/GR on periphery of cone at each 120 degrees

(c) 96 to 305 mc. (TN–17/APR–4) Antenna AT–292/URM–29, with 1 Mast Section AB–21/GR on periphery of cone at each 120 degrees

(d) 306 to 1000 mc. (TN–187/URM–29) Antenna AT–292/URM–29, without Mast Sections AB–21/GR

The antenna shall be tilted so that an extension of the ground plane intersects the near edge of the engine compartment at its mid-point.

4.3.1.3 *Radiation test, electrical accessory units (tactical vehicles).*—The radio interference radiated from all electrical accessory units submitted for test in accordance with 3.7.4 shall be measured over the frequency range of .15 to 1000 mc. The accessories shall be tested ungrounded, and powered by their standard power supply.

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4.3.1.3.1 *Test Set AN/URM–3.*—Test Set AN/URM–3 shall be arranged with the center of horizontal 9 foot rod antenna located over the mid-point of the sub-assembly and separated 12 inches therefrom for radiation tests from .15 to 40 mc. The limits specified in table I apply.

4.3.1.3.2 *R.f. interference test set AN/URM–29.*—For radiation tests from 40 to 1000 mc. the apex of the cone of the antenna shall be located 18 inches from the nearest part of the accessory under test, and the antenna shall be so oriented that a line from the apex of the cone to the center of the accessory bisects the angle between the cone and the plane. The tests shall be made with the antenna configurations as described for the various frequency bands in 4.3.1.2. The test limits specified in table I apply.

4.3.2 *Inside radiation test for turreted vehicles (tactical).*—In addition to the outside radiation test, all turreted type tactical vehicles shall have an inside radiation test conducted over the frequency range of 1.5 to 1000 mc. Inside radiation tests shall be conducted with all electrical equipment in the vehicle in operation. The test limits specified in table II apply.

4.3.2.1 *Test Set AN/URM–3.* — Test Set AN/URM–3 shall be placed on an insulating board on the outside of the vehicle as close as possible to each antenna outlet normally provided in the vehicle. A 9 foot length of antenna lead-in Wire (W–128) shall be routed successfully in each lateral direction around the inside of the vehicle as shown by routings A and B on figure 9, and connected to Test Set AN/URM–3 with the shortest possible length of antenna wire external to the vehicle. The inside radiation tests shall be conducted with all hatches closed in order to minimize pick-up of ambient noise. The radio interference shall be measured for both routings of the antenna wire at each antenna outlet location over the frequency range of 1.5 to 40 mc.

4.3.2.2 *R.f. interference test Set AN/URM–29.*—The antenna of r.f. interference test Set AN/URM–29 shall be placed inside of the turret of the vehicle as shown on figure 10, and the radio interference measured inside the vehicle over the frequency range of 40 to 1000 mc. The antenna configuration used for the various bands shall be as follows:

(a) 40 to 305 mc. Antenna AT–292/URM–29, with 1 Mast Section AB–21/GR on periphery of cone at each 120 degrees.

(b) 306 to 1000 mc. Antenna AT–292/URM–29, without Mast Sections AB–21/GR

The plane shall be placed in a horizontal position approximately 6 inches from inside top of turret for all inside radiation tests. The limits of Table II specified apply.

4.3.3 *Conducted interference test (tactical vehicles).* — This test will be made on all types of tactical vehicles designed for installation of communication equipment, and on electrical accessories for such vehicles, over the frequency range of 1.5 to 40 megacycles. The test limits specified in table III apply. The conducted interference level shall be measured with Test Set AN/URM–3, with the required coupling network connected successively between each pair of terminals intended to supply power to communications equipment and between each such terminal and ground, at each electrical terminal box, contact block or connector.

4.3.4 *Pilot tests on nontactical vehicles.*—Pilot tests on nontactical vehicles will be run by the Government only in cases covered by 3.4.1. (See 6.12.)

4.4 **Tests of electrical accessory units.**—The conducted interference level of electrical accessories being tested under 3.7.4 shall be measured with Test Set AN/URM–3 with the required coupling network connected successfully across the terminals, and between each terminal and ground, at the power input

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terminals of the accessory. The limits specified in table III apply.

4.5 Government production testing.

4.5.1 *Tactical vehicles.* — All production tactical vehicles shall be subject to inspection by the Government in accordance with 4.5.1.1 through 4.5.1.8.

4.5.1.1 *Field test.* — The contractor shall submit, at an electrical-interference-free area approved by the contracting officer, tactical vehicles required by the Government for test, to determine that such vehicles are suppressed in accordance with this specification, and to determine the thoroughness of the visual inspection. The field test shall be conducted by authorized representatives of the contracting officer, using the equipment and procedure specified for pilot model tests.

4.5.1.2 *Initial inspection.*—An initial group of not less than 5 nor more than 25 vehicles from the first production of each type of vehicle, shall be subjected by the Government to radio interference field tests in accordance with 4.5.1.1 to determine whether the production application of the suppression system is satisfactory. If it is unsatisfactory, the contractor shall repair the rejected vehicles, and shall change his production methods to correct the cause of failure. The contracting officer may then require the testing of such additional quantity of vehicles as is considered necessary to determine that the corrected production application of the suppression system is satisfactory.

4.5.1.2.1 When a group of vehicles is submitted for the initial inspection it shall be accompanied by two copies of a list of the required and/or authorized changes of the electrical and suppression systems from the approved experimental pilot model, marked with the date of submission of the vehicles, contract number, name and address of the contractor, the nomenclature of the vehicles, and the serial number and U. S. registration number of each vehicle submitted for test.

4.5.1.3 *Production sampling of tactical vehicles.*—After vehicles have satisfactorily passed the initial inspection specified in 4.5.1.1 to 4. 5. 1. 2. 1, inclusive, the quality of the suppression system on all additional vehicles shall be controlled by subjecting random vehicles to similar tests. Whenever required by the contracting officer, a maximum of 1 percent of the vehicles shall be selected for the test. If less than 100 vehicles are being procured, at least one vehicle shall be tested after the initial inspection. If any vehicle fails to pass the test the contractor shall change his production methods to correct the cause of the failure. In addition, the contracting officer shall require the testing of such additional vehicles as is considered necessary to insure that satisfactory vehicles will be provided.

4.5.1.4 *Condition of vehicles to be tested.*— No experimental pilot model or production vehicle shall be submitted for test of radio interference unless reasonably dry and free from rain, ice, and snow. The expression "reasonably dry" shall be interpreted to mean that the vehicle shall not be visibly wet.

4.5.1.5 *Electrical subassemblies and accessories.*—No Government production testing of the vehicular electrical subassemblies or accessories, separate from the vehicle, is required.

4.5.1.6 *Incompletely assembled vehicles.*— Where vehicles are shipped in other than a completely assembled condition the required field tests shall be made on vehicles which have been assembled for road test.

4.5.1.7 *Integral suppression components.*— A spot check of integral suppression components by visual or electrical test may be required, at the discretion of the contracting officer.

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4.5.1.8 *Visual inspection of suppression system.*—In each vehicle, a thorough visual inspection shall be made of the complete installation of the suppression system, except integral suppression components, for proper mechanical and electrical connection of all suppression components, for proper mechanical and electrical connection of all suppression components such as bonds, bypass capacitors, resistor-suppressors, filters, etc., to determine conformance to the pilot model report (3.7.3). Where vehicles are shipped in other than a completely assembled condition this inspection may be made on the partially assembled vehicles. This inspection shall be performed at a point or points in the production line determined by the contracting officer. Until the initial inspection (4.5.1.1 to 4.5.1.2.1, inclusive) is completed, acceptance shall be based on compliance with the pilot model report. After completion of the initial inspection, acceptance shall be based on compliance with instructions issued by the contracting officer, based on results of the tests.

4.5.2 *Inspection, nontactical vehicles.* — Production models of all nontactical vehicles shall be subjected to a visual inspection for compliance with this specification.

4.5.3 *Check for use of Government-inspected suppression components.*—Inspection shall insure that all suppression components covered by 3.5 furnished on production vehicles, both tactical and nontactical, have been subjected to tests as required by the subsidiary specifications listed in 2.1. Inspection shall also insure that suppression components covered by 3.5 have been subjected to tests as required by the contracting officer. Where the tests of suppression components are performed at a sub-contractor's plant, inspection at the plant of the vehicle manufacturer shall determine that the components, or packages thereof, bear the stamp of the Government inspector.

4.6 **Corrosion test.** — Parts which must meet the requirements of 3.11.3 shall be tested as follows: Just prior to testing, wash the parts in petroleum ether, then in alcohol, and dry. Subject the parts to a salt spray for 200 hours at a temperature of 95 ±4° F., using a 20 percent (by weight) aqueous solution of sodium chloride. Conduct the test in a closed tank of material nonreactive to the salt spray. The spray shall be finely divided. The tank shall be so designed that there will be no direct impinging or dripping of spray upon the parts. The spray shall circulate freely about all the parts to the same degree, and no liquid which has come in contact with the parts shall return to the aspirator to be resprayed. (See 6.2 to 6.3, inclusive.)

5. PREPARATION FOR DELIVERY

5.1 The applicable section of the equipment specification shall apply.

6. NOTES

6.1 **Type approval.**—Current Army Qualified Products Lists (AQPL's) of type-approved suppression capacitors, filters and resistor-suppressors may be obtained by contracting officers from the Signal Corps Procurement Agency, 225 South 18th Street, Philadelphia 3, Pa., by referring to the Specification Number covering the particular component. The AQPL's should be distributed by the contracting officer to contractors or prospective contractors. Indication of approval of any item, by its inclusion in an AQPL or by other means, as being suitable for suppression purposes and in accordance with the applicable specification, covers use of the component only for suppression purposes. Current QPL's of type-approved JAN capacitors are also available from the same source; however, see 3.5.2.

6.2 **Contractual samples.** — The samples mentioned in 3.5.1 through 3.5.4.3, and 3.11.3 should be submitted as described in 6.2.1. Each sample should be securely tagged with sufficient identifying information such as name and address of the manufacturer of

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the component, his type or model number for the components, the name and the address of the prospective bidder or contractor on the order for the equipment, the order and contract number pertaining to the procurement of the equipment, and the notation "CONTRACTUAL SAMPLE." Such samples should be accompanied by a statement from the contractor giving all pertinent information concerning the reason for requesting that the samples be inspected and tested, and providing engineering information and wiring diagrams concerning the proposed application of the items represented by the samples.

6.2.1 Contractual samples received by the contracting officer should be submitted to the Signal Property Officer, Bldg. 41, Supply Division, Evans Signal Laboratory, Belmar, N. J., Attn: Suppression and General Engineering Branch, Coles Signal Laboratory.

6.3 Noncontractual samples.—Noncontractual samples of suppression components (those not submitted in accordance with 3.5.1 through 3.5.4.3, and 3.11.3) may be submitted to the address specified as given in paragraph 6.2.1 for inspection and test. The samples should be accompanied by a statement from the Government agency concerned, giving all pertinent information concerning the reasons for requesting that they be tested and their proposed application. Each sample should be securely tagged with sufficient identifying information such as name and address of the manufacturer and his type or model number, and the notation "NONCONTRACTUAL SAMPLE." The samples should be accompanied by two copies of a statement of release from obligation, as indicated below:

Number of Samples:_____Type_____
Remarks: _____

The undersigned distinctly understands and agrees that the sample(s) itemized above are offered the Government free of charge and without any obligation whatsoever on the part of the Government, either expressed or implied. It is further understood that the Government shall be under no obligation to test the samples, to furnish results of the tests, enter into any correspondence concerning the sample(s), or to return the sample(s). The undersigned state that it is their standard and commercial practice to furnish such sample(s), in the quantity listed above, to the trade without charge.

_____
(Signature)

_____
(Firm Name)

_____
(Date)

6.4 Testing of pilot model and production vehicles.—Test equipment and personnel for conducting the inspection and test of the suppression systems on pilot models, and on production vehicles undergoing the Government field tests, will be provided by Coles Signal Laboratory. Contracting officers should notify the Director, Coles Signal Laboratory, Fort Monmouth, New Jersey *at least 10 days* before a pilot model will be available for inspection and test of the suppression system. Results of the tests, and a report covering the method of suppression to be followed on any subsequent pilot model and/or production unit will be furnished to the contracting officer. The report should be included as part of any approval of the pilot model to which it applies.

6.5 Representatives of the contracting officer.—The contracting officer represents the U. S. Government in all dealings with a contractor who is furnishing equipment that is required to be suppressed in accordance with this specification, and in all direct dealings with the manufacturer (sub-contractor) who is furnishing suppression components or electrical components or accessories to the contractor for installation on the equipment. It is to be understood that any Government agency or representative can perform those functions of the contracting officer that are mentioned in this specification, to the extent

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that the contracting officer has specifically delegated his authority to such Government agency or representative. It is to be further understood that Government inspectors represent the contracting officer for supervision of inspection and testing of equipment and the components and parts thereof, at the contractor's or manufacturer's plant.

6.5.1 The proposal request and contract will usually delegate the following functions to a Government engineering agency:

(1) Approval for the use of capacitors or filters under conditions that are more severe than the one specified in the applicable specification therefor.

(2) Testing of capacitors or filters that do not conform to the applicable specification to determine whether such components are acceptable for the proposed application.

(3) Inspection, testing, and approval of samples of suppression components to be approved before being furnished as part of the equipment on order.

(4) Inspection, testing, and approval of the suppression system on the pilot model.

(5) Use and maintenance of Government test equipment.

(6) The Government agency designated to perform the above functions will usually be Coles Signal Laboratory, Fort Monmouth, New Jersey.

6.6 Proposal request and contract.

6.6.1 *General information.*—The following general information is given for the guidance of the contracting officer:

6.6.2 *Inspection of the suppression system.* —Government inspectors representing the procuring service concerned should be responsible for supervision of the visual inspection of the suppression system on production units. Assistance from representatives of Coles Signal Laboratory may be obtained for the following:

(1) Approval of interference-free area.

(2) Use and maintenance of Government test equipment for testing production models.

6.6.3 *Responsibility for testing of samples and pilot models.* — It is recommended that the contracting officer delegate the testing of the suppression system on the pilot models, the testing of samples submitted in accordance with 3.5.1 through 3.5.4.3 of this specification, and the testing of samples of other items relating to suppression of the vehicle, to the Coles Signal Laboratory.

6.6.4 *Recommendations.*

6.6.4.1 The contracting officer should direct that all drawings, prints, lists of suppression components, etc., that are required by this specification to be provided by the contractor, be sent to the Coles Signal Laboratory for information and necessary action.

6.6.4.2 If requested by the contractor, information covering test procedure and Government test equipment may be obtained by the contracting officer upon application to the Coles Signal Laboratory.

6.6.4.3 Close coordination should be maintained among the contracting officer, the inspection agency, and the Coles Signal Laboratory to insure that any changes in pilot models or production items that might affect radio interference suppression will be satisfactorily provided for by necessary changes in the suppression system.

6.7 Sources of electrical interferences.— The more common sources and emanators of interference are listed below, but should not be considered as the only possible sources of interference:

a. Ignition system of the engine:

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(1) High-tension ignition circuit, including all high-voltage wiring of the ignition system, the ignition coil or magneto, distributor and spark plugs.

(2) Low-tension ignition system, including low-voltage wiring of the ignition system, ignition coil, ignition switch and breaker points.

b. Charging circuit, including the generator, regulator, and associated wiring.

c. Low-voltage electrical system, which includes all wiring for the electrical subassemblies and accessories of the vehicles, such as:

(1) Electric fans, windshield wipers, indicating devices, traversing equipment, gyrostabilizer equipment, etc.

(2) Auxiliary engine-generator units.

d. Intermittent contact between parts.

e. Electrostatic discharges from track systems of track-laying vehicles, nondriven wheels, and fan belts.

6.8 Contractor's tests on pilot model.—The contractor, in making the tests required by 3.7.1 on the pilot model, may use commercial radio noise meters or sensitive radio receivers covering the required frequency range; however, the fact that a vehicle may be suppressed to the point of emanating no interference detectable on such equipment does not necessarily indicate that it will be satisfactory when tested with the Government's test equipment.

6.9 Tests and consultation on suppression of automotive electrical subassemblies.—Automotive electrical subassemblies, as defined in 3.1.5, may be submitted by the contractor or sub-contractor, for qualitative tests of the suppression characteristics and assistance in the design of suitable suppression systems, to Signal Property Officer, Building 41, Supply Division, Evans Signal Laboratory, Belmar, N. J. Attn: Suppression and General Engineering Branch, Coles Signal Laboratory. Consultation on design characteristics of electrical subassemblies in the development stages, and on integral suppression components for inclusion therein, to facilitate suppression over the frequency range required by this specification, can be obtained from Coles Signal Laboratory, Fort Monmouth, New Jersey.

6.10 Tactical vehicle suppression system.—Signal Corps Drawing SC–D–28120 "Typical Tactical Vehicle Suppression System," is available for the information of the contractor; however, the use of a system such as shown therein does not assure that the vehicle to which it is applied will meet the requirements of this specification.

6.11 Railroad equipment. — It is recommended that the contracting officer furnish complete information regarding the nature and intended use of railroad equipment to be suppressed, to Coles Signal Laboratory, Fort Monmouth, N. J. in order to obtain information regarding the applicable paragraphs of this specification, for inclusion in the bid request and contract.

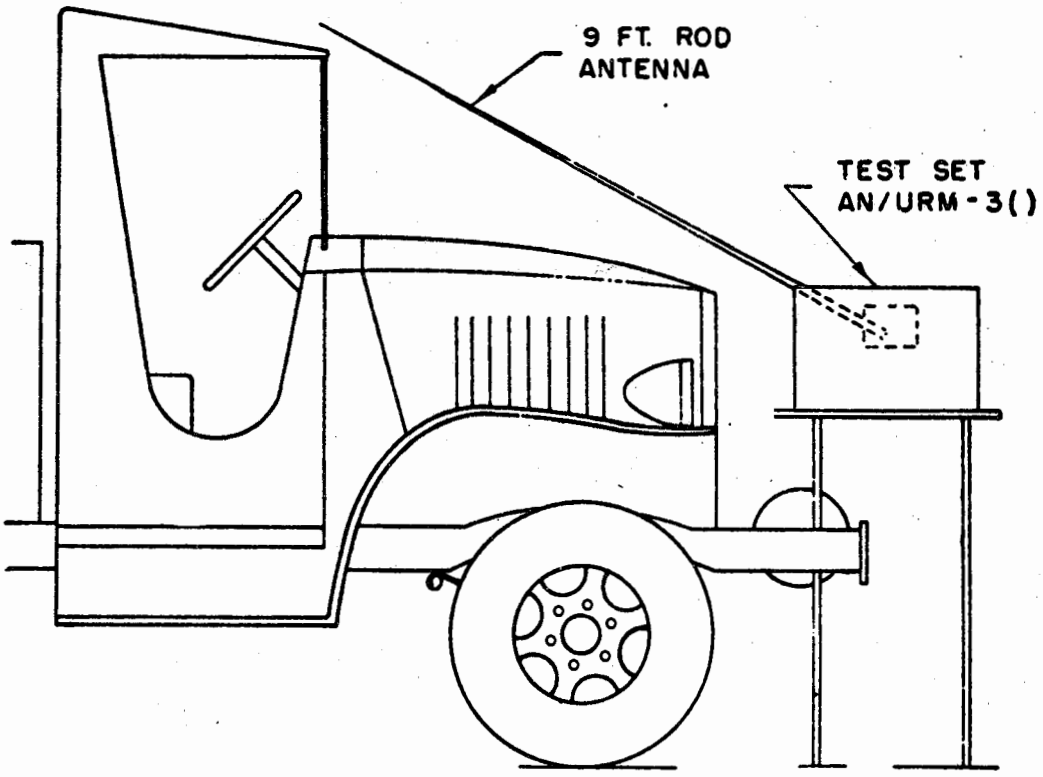
6.12 Information on the test equipment and the manner in which it will be used for the testing of non-tactical vehicles may be obtained from Coles Signal Laboratory, Fort Monmouth, N. J.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Army—Signal Corps

Other interest:
Army—EOT
Navy—ShAMCY
Air Force.

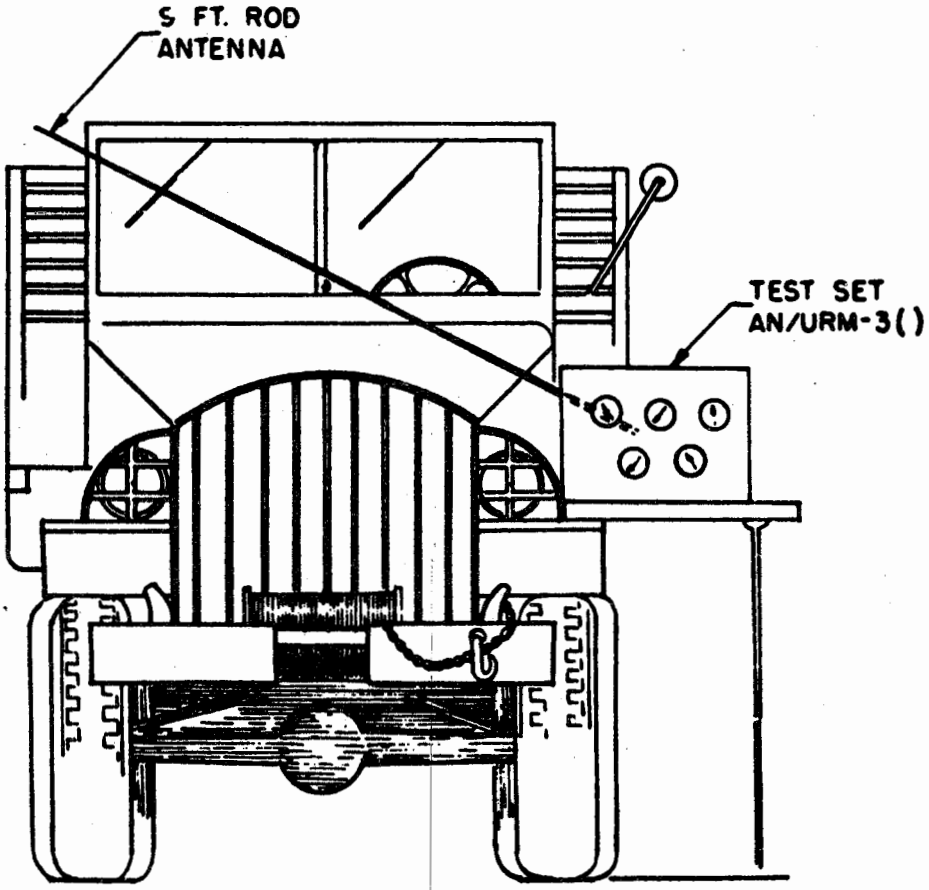
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NOTE
1. THE TEST SET SHALL BE LOCATED ON THE LONGITUDINAL AXIS OF THE VEHICLE.
2. THE TOP OF THE TEST SET SHALL BE IN THE SAME PLANE AS THE TOP OF THE RADIATOR GRILL.
3. THE TEST SET SHALL BE LOCATED AS CLOSE AS POSSIBLE TO THE VEHICLE WITHOUT THE CENTER OF THE ANTENNA EXCEEDING TWO FEET FROM TOP OF ENGINE COMPARTMENT OR APPROACHING CLOSER THAN ONE FOOT THERETO.

FIGURE 1.—*Antenna position for trucks front radiation test, 0.15–40 mc.*

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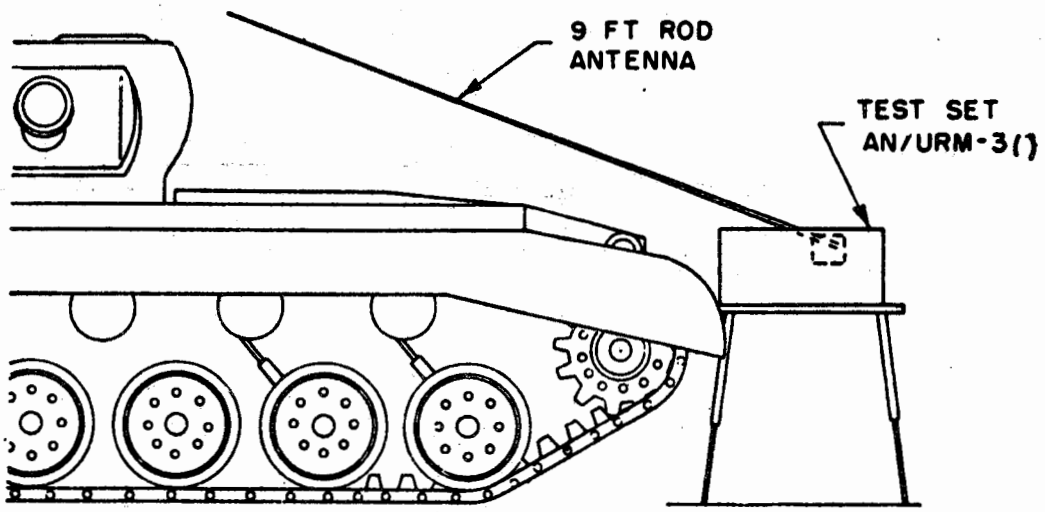


NOTE:

1. THE TEST SET SHALL BE INSULATED FROM THE BODY OF THE VEHICLE WITH A NON-CONDUCTING MATERIAL.
2. THE ROD ANTENNA SHALL BE LOCATED AS CLOSE AS POSSIBLE TO THE LATERAL AXIS OF THE ENGINE.
3. THE TIP OF THE ANTENNA SHALL BE LOCATED IN A VERTICAL PLANE THROUGH THE EDGE OF THE OPPOSITE FENDER OR SIMILAR OUTERMOST EXTREMITY OF METAL, BUT THE MIDPOINT OF THE ANTENNA SHALL NOT BE PLACED CLOSER THAN ONE FOOT FROM THE TOP OF THE ENGINE COMPARTMENT FOR ANY VEHICLE.
4. A SIDE RADIATION TEST IS TO BE PERFORMED WITH THE TEST SET LOCATED ON BOTH RIGHT AND LEFT FENDERS.

FIGURE 2.—*Antenna position for trucks side radiation test, 0.15–40 mc.*

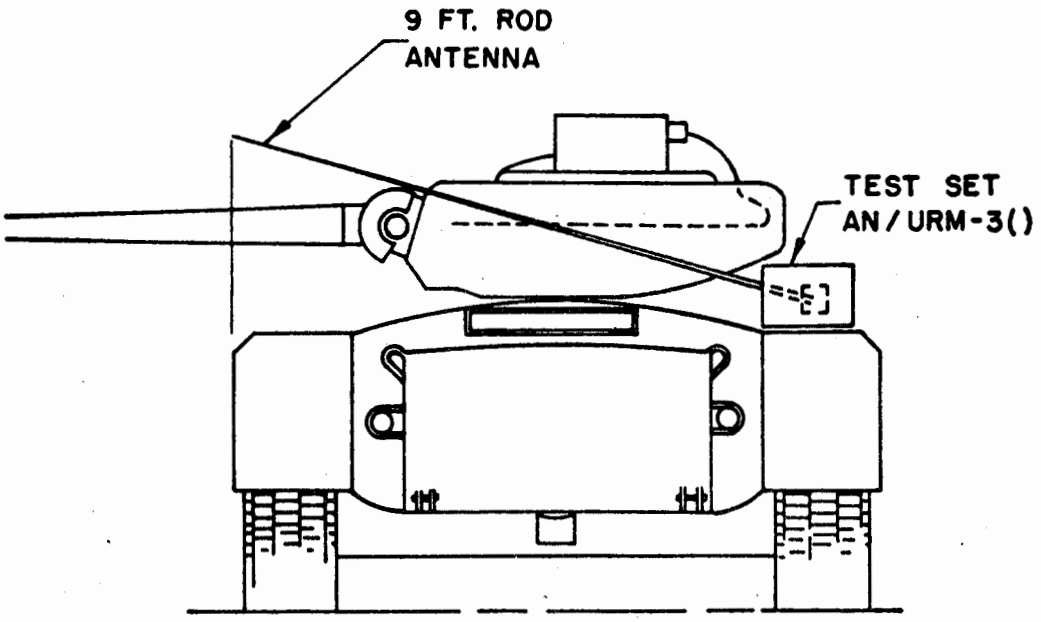
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NOTE:
1. THE ANTENNA SHALL BE PLACED ALONG THE LONGITUDINAL AXIS OF THE VEHICLE.
2. THE TOP OF THE TEST SET SHALL BE IN THE SAME PLANE AS THE LOWER REAR EDGE OF THE AIR OUTLET GRILL.
3. ALL TESTS SHALL BE CONDUCTED WITH THE GUN ROTATED PERPENDICULAR TO THE LONGITUDINAL AXIS OF THE VEHICLE.
4. THE TEST SET SHALL BE LOCATED AS CLOSE AS POSSIBLE TO THE VEHICLE WITHOUT THE MID-POINT OF THE ROD ANTENNA EXCEEDING TWO FEET FROM TOP OF ENGINE COMPARTMENT OR APPROACHING CLOSER THAN ONE FOOT THERETO.

FIGURE 3.—*Antenna position for tanks and tank type vehicles end radiation test, 0.15–40 mc.*

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NOTE:

1. THE TEST SET SHALL BE INSULATED FROM THE BODY OF THE VEHICLE WITH A NON-CONDUCTING MATERIAL.
2. GUN SHALL BE ROTATED PERPENDICULAR TO THE LONGITUDINAL AXIS OF THE VEHICLE.
3. THE ROD ANTENNA SHALL BE LOCATED AS CLOSE TO THE LATERAL AXIS OF THE ENGINE AS POSSIBLE.
4. THE TIP OF THE ANTENNA SHALL BE LOCATED IN A VERTICAL PLANE THROUGH THE EDGE OF THE OF THE OPPOSITE FENDER OR SIMILAR OUTERMOST EXTREMITY OF METAL, BUT THE MIDPOINT OF THE ANTENNA SHALL NOT BE PLACED CLOSER THAN ONE FOOT FROM THE TOP OF THE ENGINE COMPARTMENT FOR ANY VEHICLE.
5. A SIDE RADIATION TEST IS TO BE PERFORMED WITH THE TEST SET LOCATED ON BOTH THE RIGHT AND LEFT FENDER.

FIGURE 4.—*Antenna position for tanks and tank type vehicles—radiation side radiation, 0.15–40 mc.*

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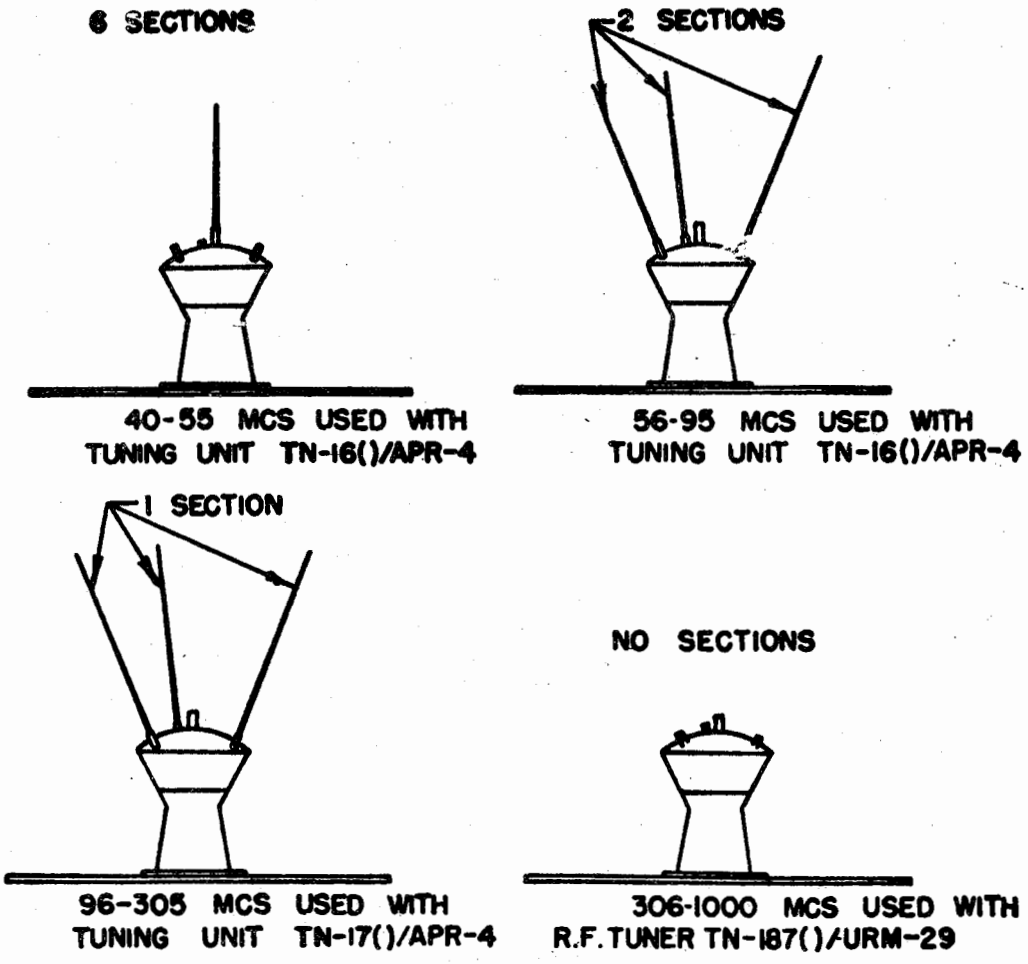
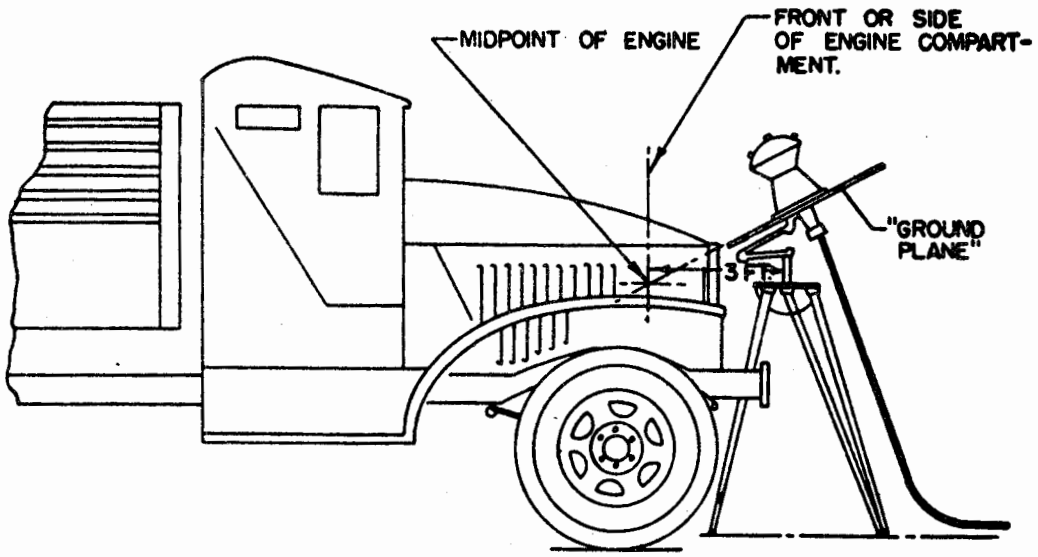


FIGURE 5.—*Antenna AT-892( )/URM-29 and mast sections AB-81/GR for outside radiation test 40-1000 mc.*

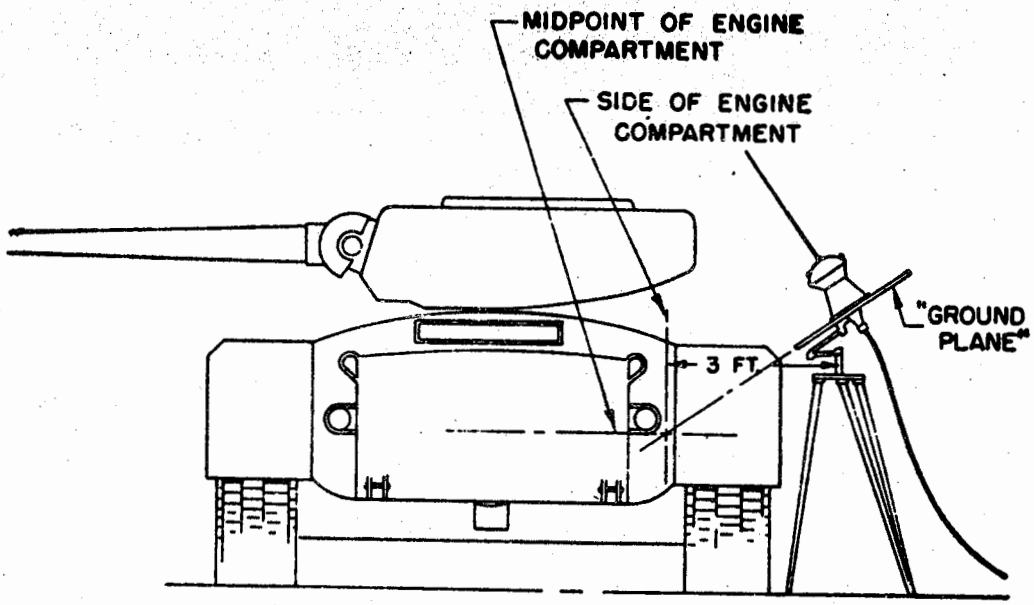
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NOTE:
1. RADIATION TESTS ARE REQUIRED ON THREE SIDES OF EACH TRUCK OVER FREQUENCY RANGE 40 TO 1000 MC WITH SAME ANTENNA AND "GROUND PLANE" POSITIONING.
2. PRIOR TO TILTING, "GROUND PLANE" SHALL BE IN THE SAME HORIZONTAL PLANE AS TOP OF RADIATOR.
3. ANTENNA TILTED SO THAT EXTENSION OF "GROUND PLANE" INTERSECTS FRONT OR SIDES OF ENGINE COMPARTMENT AT MIDPOINT.

FIGURE 6.—*Antenna positioning of test Set AN/URM-29( ) in radiation test of trucks 40-1000 mc.*

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NOTE:

1 GUN SHALL BE ROTATED PERPENDICULAR TO THE LONGITUDINAL AXIS OF THE VEHICLE.

2. THE ANTENNA SHALL BE LOCATED AS CLOSE AS POSSIBLE TO THE LATERAL AXIS OF THE ENGINE.

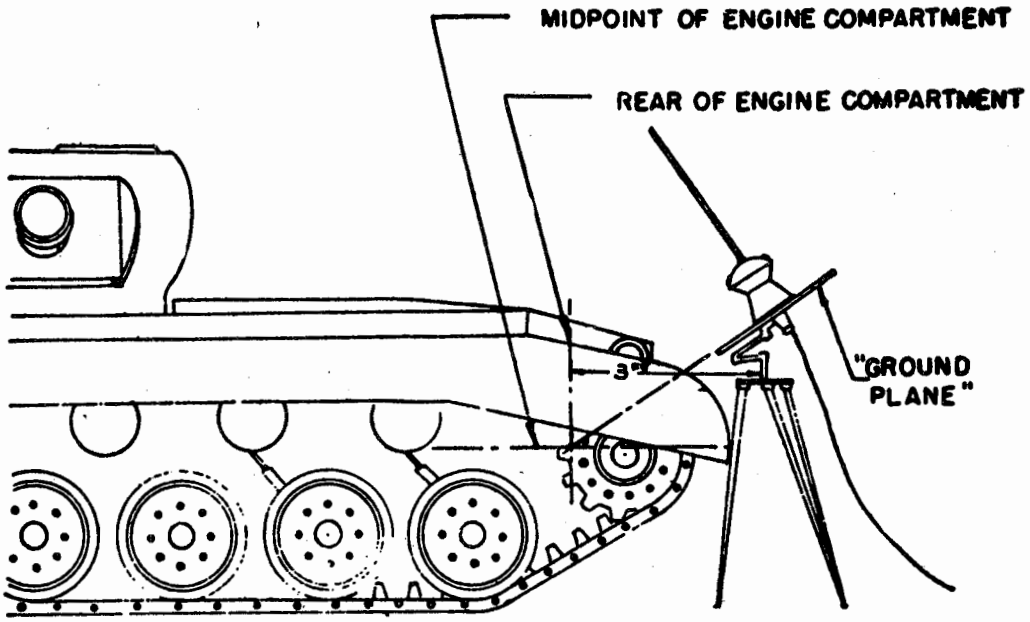
3. PRIOR TO TILTING,"GROUND PLANE" SHALL BE IN THE SAME HORIZONTAL PLANE AS THE LOWER EDGE OF THE AIR OUTLET GRILL.

4. ANTENNA TILTED SO THAT EXTENSION OF "GROUND PLANE" INTERSECTS NEAR SIDE OF ENGINE COMPARTMENT AT VERTICAL MIDPOINT.

5 A SIDE RADIATION TEST IS TO BE PERFORMED WITH THE ANTENNA LOCATED ON EACH SIDE, ON THE LATERAL AXIS OF THE ENGINE.

FIGURE 7.—*Antenna position for tanks and tank type vehicles—radiation 40–1000 mc.*

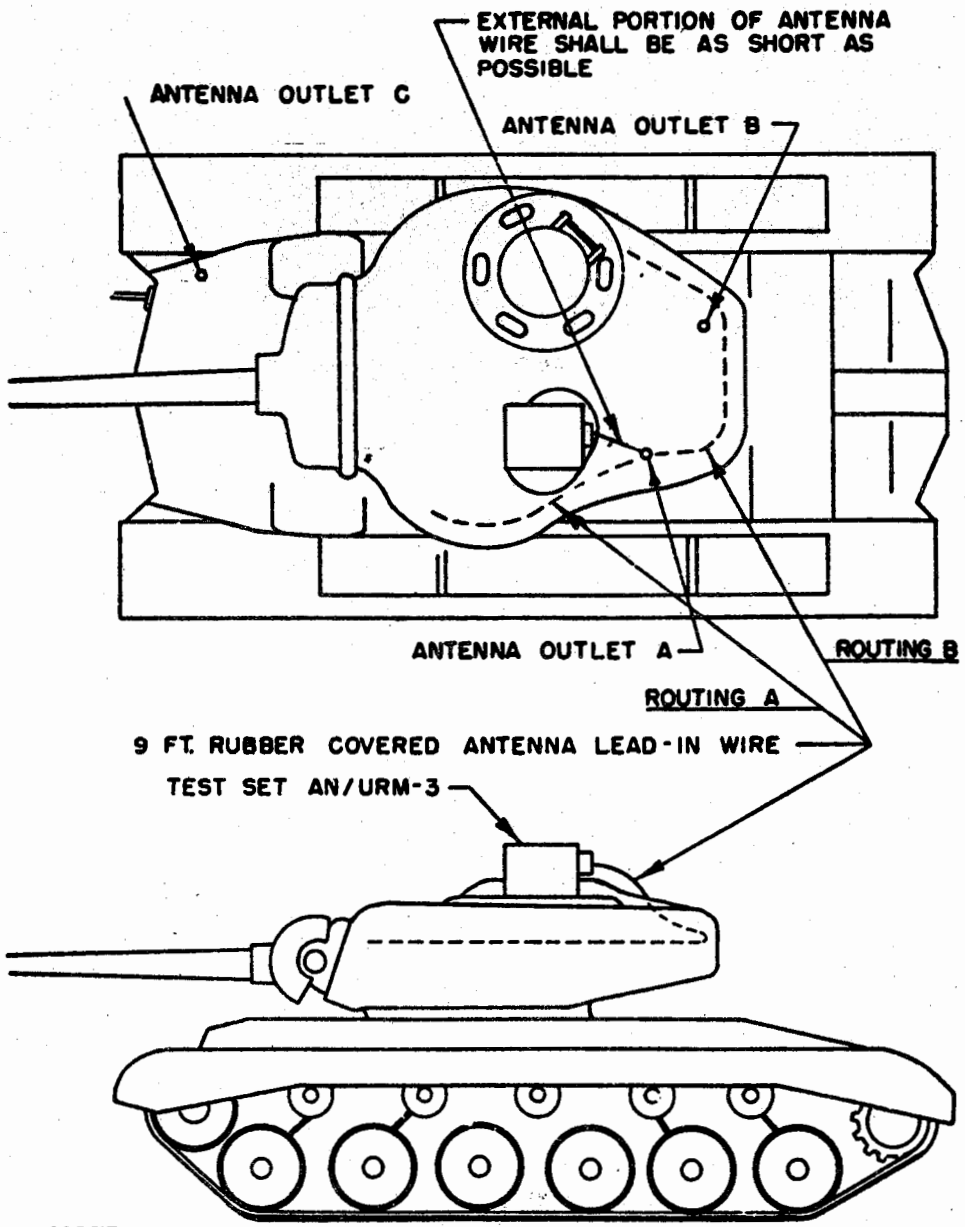
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NOTE
1. THE TEST SHALL BE CONDUCTED WITH THE GUN ROTATED PERPENDICULAR TO THE LONGITUDINAL AXIS OF THE VEHICLE
2. THE ANTENNA SHALL BE PLACED ALONG THE LONGITUDINAL AXIS OF THE VEHICLE.
3. PRIOR TO TILTING, "GROUND PLANE" SHALL BE IN THE SAME HORIZONAL PLANE AS THE LOWER EDGE OF THE AIR OUTLET GRILL.
4. ANTENNA TILTED SO THAT EXTENSION OF "GROUND PLANE" INTERSECTS REAR OF ENGINE COMPARTMENT AT MIDPOINT.

FIGURE 8.—*Antenna position for tanks and tank type vehicles radiation, 40–1000 mc.*

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NOTE:

1. THE TEST SET SHALL BE INSULATED FROM THE BODY OF THE VEHICLE WITH A NON-CONDUCTING MATERIAL.

2. A RADIATION TEST IS REQUIRED FOR BOTH ANTENNA ROUTINGS AT EACH ANTENNA OUTLET ON THE VEHICLE.

FIGURE 9.—*Antenna positioning for inside radiation test 0.15–40 mc.*

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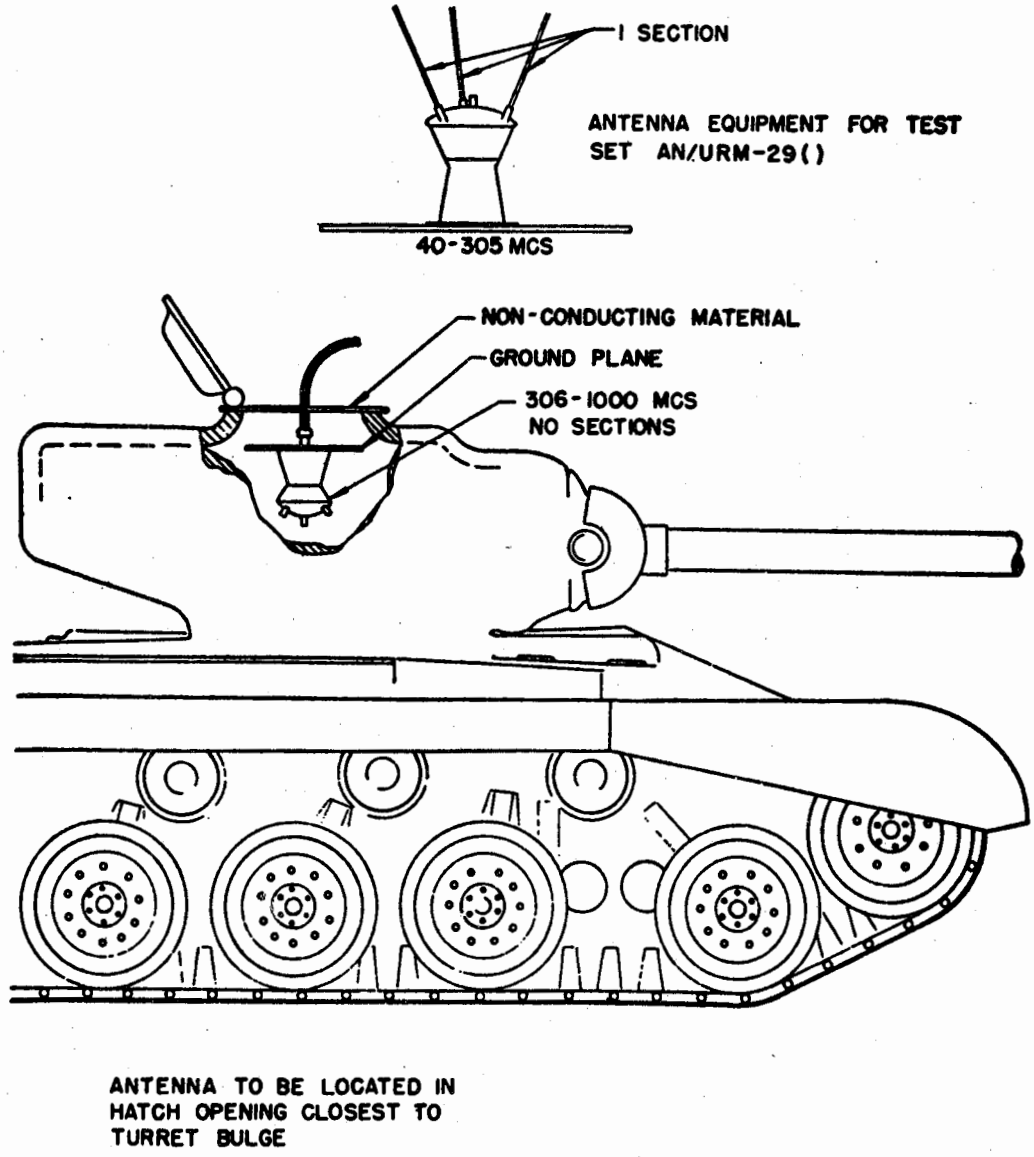


FIGURE 10.—*Antenna positioning of test Set AN/URM-29( ) for inside radiation test, 40 to 1000 mcs.*

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